

FROM RELIABLE TEXT TO REAL VOICES: TRUST-AWARE PROGRESSIVE ADAPTATION FOR LOW-RESOURCE TTS

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ABSTRACT

Adapting multilingual text-to-speech (TTS) models to low-resource languages requires learning target-language pronunciation while retaining zero-shot voice cloning with limited transcribed speech. Synthetic speech and real recordings with automatic speech recognition (ASR) pseudo-labels offer scalable supervision, but differ in transcript reliability and acoustic diversity. We find that their training order affects the balance between content accuracy and speaker similarity, while noisy pseudo-labels can undermine pronunciation learned from synthetic speech. We propose synthetic-to-real adaptation with transcript-agreement weighting: synthetic pairs first establish text–speech correspondences, followed by real recordings to restore speaker conditioning, with agreement between two ASR systems determining each example’s loss weight. Experiments with FireRedTTS3 on Burmese and Lao, and OmniVoice on Burmese, show improved content accuracy and naturalness with competitive speaker similarity. Jointly considering supervision order and reliability when combining synthetic and real speech offers a practical path to low-resource TTS adaptation with less manual transcription.

Index Terms— Low-resource TTS, synthetic-to-real adaptation, pseudo-label reliability, zero-shot voice cloning

1. INTRODUCTION

Multilingual text-to-speech (TTS) has advanced from reproducing fixed training speakers to cloning unseen voices from short reference recordings [1, 2, 3]. Recent systems further broaden language coverage, with OmniVoice extending zero-shot voice cloning to over 600 languages. However, this expansion still faces imbalanced training data and inconsistent transcript and acoustic quality [4]. For low-resource adaptation, the challenge is therefore not simply to add a language, but to learn its pronunciation while retaining reference-speaker control from limited supervision [5].

To reduce this dependence on transcribed speech, two supervision sources are available: synthetic pairs and real recordings with ASR pseudo-labels. Synthetic pairs can expand target-language text coverage, as demonstrated by synthetic pretraining followed by target-speaker adaptation [6]. Yet synthetic augmentation can reduce speaker similarity, motivating domain-conditioned training in ZeSTA [7]. Real recordings offer another route: CosyVoice 2 constructs multilingual training data through ASR pseudo-labeling and quality filtering [8]. Together, these studies motivate using both sources, but also expose distinct limitations: synthetic supervision is constrained by teacher voices, whereas real supervision depends on transcript reliability.

These limitations raise a question for language adaptation: how should the two sources be combined to improve pronunciation without sacrificing zero-shot voice cloning? In our experiments, synthetic adaptation improves content accuracy but can reduce speaker similarity; subsequent real-speech adaptation restores similarity, while noisy pseudo-labels can erode the content gains. Reversing the order produces a different trade-off. Thus, beyond the speaker-personalization setting of prior synthetic augmentation work, we examine how supervision order and pseudo-label reliability jointly affect language adaptation and reference-speaker control.

Guided by these observations, we propose trust-aware progressive adaptation. S-stage adaptation first learns target-language text–speech correspondences from synthetic pairs. R-stage adaptation then uses real recordings to recover speaker conditioning, with transcript agreement between two ASR systems serving as a proxy for pseudo-label reliability. The primary transcript remains the training text; agreement determines each example’s loss weight. We evaluate this design with FireRedTTS3 [9] on Burmese and Lao and OmniVoice on Burmese, alongside controlled ablations and reference-conditioning and acoustic-history interventions.

Our contributions are:

- **Analysis.** We characterize how supervision order affects content accuracy and speaker similarity.
- **Method.** We propose synthetic-to-real adaptation with transcript-agreement weighting for noisy pseudo-labels.
- **Validation.** We demonstrate effectiveness across languages and backbones, supported by controlled interventions.

2. METHOD

Figure 1 presents trust-aware progressive adaptation: S-stage adaptation learns text–speech correspondences, followed by R-stage adaptation with transcript-agreement weighting to recover reference-speaker control. The Base model, Synthetic-Stage-Model, and Final-Stage-Model are abbreviated Base, S, and S→R.

2.1. Synthetic supervision

The fixed Teacher TTS T maps target-language text x_i and teacher voice v_i to speech $y_i^S = T(x_i, v_i)$. Quality filtering (Section 3) and same-voice reference pairing yield $\mathcal{D}_S = \{(x_i, y_i^S, r_i^S)\}$, where r_i^S is reference speech, i indexes examples, and superscripts S/R denote synthetic/real data. S-stage adaptation uses unit weights and the original synthesis inputs as training text; filtering does not guarantee error-free pronunciation.

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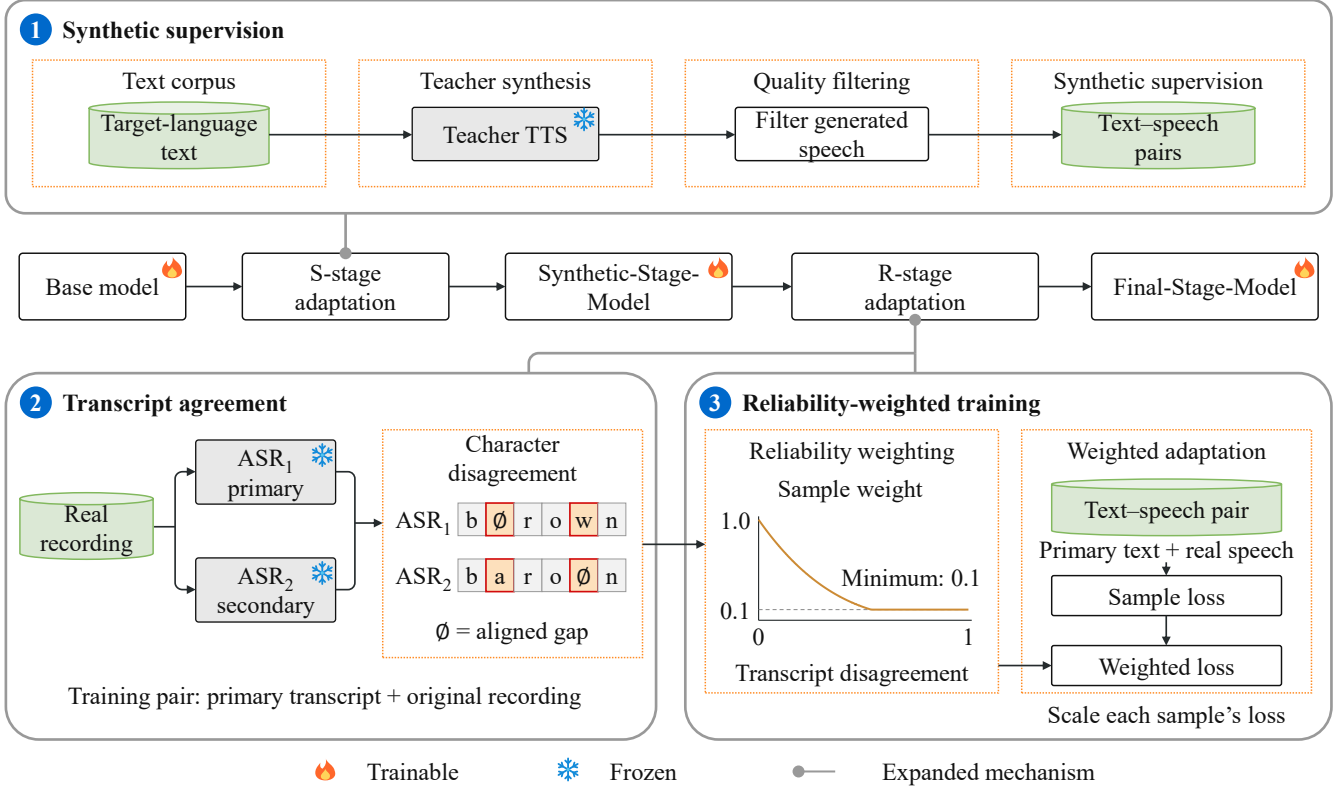


Fig. 1. Trust-aware progressive adaptation. Synthetic supervision establishes text-speech correspondences in S-stage adaptation; transcript agreement guides reliability-weighted training on real recordings in R-stage adaptation. The primary transcript remains the training text.

2.2. Transcript agreement

For real recording y_i^R , fixed ASR₁ (primary), denoted A_1 , supplies $\tilde{x}_i = A_1(y_i^R)$; fixed ASR₂ (secondary), denoted A_2 , supplies only a comparison transcript. The training set is $\mathcal{D}_R = \{(\tilde{x}_i, y_i^R, r_i^R)\}$, without transcript fusion or correction. Reference speech r_i^R is a distinct 1–10 s utterance from the same local speaker within a recording. Controls fix pairings and reference text, exclude ineligible targets, and pair mixture references within each domain.

Normalized character disagreement measures transcript agreement:

$$d_i = \frac{\text{ED}(\mathcal{N}(\tilde{x}_i), \mathcal{N}(A_2(y_i^R)))}{\max(1, |\mathcal{N}(\tilde{x}_i)|)}. \quad (1)$$

ED and $|\cdot|$ denote code-point Levenshtein distance and length. $\mathcal{N}$ applies Unicode NFC and removes whitespace and punctuation/symbol/control categories, retaining letters, digits, and combining marks without case, numeral, or Zawgyi conversion. This normalization is for scoring, not training-text construction. Empty primary labels are excluded; valid empty secondary hypotheses give $d_i = 1$, distinct from ASR request failures.

Lower d_i means greater agreement, not verified correctness: both recognizers may share errors. Figure 1 labels d_i as character disagreement and transcript disagreement; its $\emptyset$ denotes an alignment gap, not an inserted training character.

2.3. Reliability-weighted training

Reliability weighting. To limit noisy supervision’s contribution [10], we assign

$$w_i^{(\gamma)} = \max(w_{\min}, [1 - \min(d_i, 1)]^\gamma), \quad (2)$$

where $\gamma > 0$ controls decay and $w_{\min}$ is the minimum weight. Default cubic weighting uses $\gamma = 3$ and $w_{\min} = 0.10$: perfect agreement receives unit weight, while the floor retains high-disagreement examples with reduced influence. Parameter selection is described in Section 3.

Weighted adaptation. For minibatch $\mathcal{B}$ with $|\mathcal{B}|$ examples and backbone sample loss $\ell_\theta(y_i^R, \tilde{x}_i, r_i^R)$ at trainable parameters θ ,

$$\mathcal{L}_R(\theta) = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} w_i^{(\gamma)} \ell_\theta(y_i^R, \tilde{x}_i, r_i^R), \quad (3)$$

averages complete weighted losses by example count, not weight sum. Weighting changes relative contributions and loss scale, but does not proportionally scale AdamW updates. Offline scoring adds no ASR computation at inference. The two-stage update is

$$\begin{aligned} \theta_S &= \mathcal{A}_{B_S}(\theta_0; \mathcal{D}_S, \mathbf{1}), \\ \theta_{SR}^{(\gamma)} &= \mathcal{A}_{B_R}(\theta_S; \mathcal{D}_R, w^{(\gamma)}), \end{aligned} \quad (4)$$

where $\mathcal{A}_B(\theta; \mathcal{D}, w)$ performs B updates from θ on $\mathcal{D}$ with weights w . θ_0 , θ_S , and $\theta_{SR}^{(\gamma)}$ denote Base, S, and S→R parameters; B_S/B_R are stage update budgets, $\mathbf{1}$ denotes unit weights, and $w^{(\gamma)}$ collects real-example weights. We retain each backbone’s architecture and objective, restarting the optimizer and schedule between stages. R adapts directly from Base; R→S reverses the stages.

Table 1. Results (CER in %). H is the equal-weight harmonic mean of character accuracy and SIM-O (0–100). MOS: naturalness (1–5). S→R: 1.0/0.5 denotes uniform weighting; bold: column best; shading: cubic; –: unavailable.

Strategy	FireRedTTS3								OmniVoice			
	Burmese				Lao				Burmese			
	CER ↓	SIM-O ↑	H ↑	MOS ↑	CER ↓	SIM-O ↑	H ↑	MOS ↑	CER ↓	SIM-O ↑	H ↑	MOS ↑
S	17.427	0.45126	58.36	3.15	13.488	0.61081	71.61	3.38	8.21	0.6864	78.54	4.23
R	54.093	0.73139	56.41	1.86	37.062	0.75372	68.60	2.12	10.74	0.7328	80.48	2.35
S→R: 1.0	23.274	0.69424	72.89	–	14.987	0.69931	76.74	–	10.21	0.7142	79.56	–
S→R: 0.5	20.914	0.69962	74.24	–	14.598	0.69677	76.74	–	8.03	0.7094	80.10	–
R→S	14.714	0.36357	50.98	2.72	10.643	0.59592	71.50	3.27	7.36	0.6948	79.41	4.32
S→R: cubic	16.902	0.69973	75.97	4.12	13.965	0.69684	77.00	3.97	6.85	0.7098	80.57	4.50

Burmese: Common400 CER and Clone300 SIM-O. Lao: FLEURS404 CER and SIM-O against one reference; CER SD across seeds is 0.031–0.119 points. Base MOS in column-group order: 2.17 / 2.54 / 4.40.

2.4. Backbone objectives and reference conditioning

FireRedTTS3 [9] updates the generative core, freezing RedAE and CAM++ [11]. Its complete loss $\ell_i = \ell_{\text{flow},i} + 0.1\ell_{\text{stop},i}$ combines flow matching [12] and stop prediction and is weighted as a whole. OmniVoice retains its masked acoustic-token prediction objective [4] with the same weighting.

FireRedTTS3 conditions on reference speech r through acoustic prompt $z^r = E_{\text{AE}}(r)$ and speaker embedding $e^r = E_{\text{spk}}(r)$, using the frozen RedAE encoder and CAM++ extractor, respectively; these encoders are omitted from Fig. 1. For text x and K latent patches,

$$p_{\theta}(z_{1:K} \mid x, z^r, e^r) = \prod_{k=1}^K p_{\theta}(z_k \mid x, z^r, e^r, z_{<k}), \quad (5)$$

Here p_{θ} is the model distribution; acoustic history $z_{<k}$ uses ground-truth patches in training, generated at inference. Frozen encoders still permit changes in core conditioning. Reference swaps and matched-prefix interventions (Section 4) diagnose these changes in FireRedTTS3; they are not adaptation stages.

3. EXPERIMENTS

3.1. Experimental setup

Data. Burmese S/R comprise 160,918 Azure Nilar/Thiha pairs (291.148 h) and 48,043 DVB pairs (92.263 h); Lao has 119,246 S and 26,004 R pairs (200.004/45.117 h). ASR₁ is Gemini 2.5 Flash; ASR₂ is Omnilingual ASR [13]. Synthetic filtering checks valid audio/text, 0.4–20 s duration, ASR CER ≤ 20%, speech fraction ≥ 0.65, clipping ≤ 0.005, deduplication, evaluation exclusions, reference eligibility, and a 1,024-token limit. Burmese retains 160,918 of 171,240 synthetic candidates; 717 real targets lack references. Real speech spans 300 recordings, 1,200 local speaker groups, and 286 unverified cross-recording identity clusters.

Training. Burmese S/R use 53,640/16,015 updates at $2 \times 10^{-6}/10^{-6}$; Lao uses 39,749/8,668 updates with matched per-domain schedules across orders. AdamW [14] uses 3% warmup and cosine decay. Weighting controls and Lao use effective batch 3; weighting controls use one device, accumulation 3, and seeds 42, 17, and 73. On Dev120 (60 texts, two references each), we search $\gamma \in \{0.5, 1, 2, 3, 4, 6\}$ and $w_{\min} \in \{0, 0.05, 0.10, 0.20\}$. Minimum three-seed mean CER subject to SIM-O ≥ 0.6970 selects $\gamma = 3$, $w_{\min} = 0.10$. Broadcast R endpoints are fixed at 16,015 updates, not test-selected.

Table 2. Burmese systems on Common400 / Clone300. *: fixed voices, with descriptive SIM-O scores.

System	CER (%) ↓	SIM-O ↑
Fish Audio S2-Pro [18]	83.894	0.6376
MMS-TTS-mya* [19]	20.487	0.1029
F5-Myanmar-TTS v2 [20]	35.241	0.5863
IMS-Toucan [21]	95.402	0.3298
mmSpeech Tacotron* [22]	78.695	0.0914
OmniVoice Base [4]	15.040	0.7021
FireRedTTS3 cubic	16.902	0.6997
OmniVoice cubic	6.850	0.7098

Evaluation. Burmese CER uses Common400 (100 general/300 cloning conditions; 150 texts), scored by Omnilingual ASR Unlimited 7B v2. Clone300 measures speaker similarity (SIM-O) using 30 Burmese/English/Chinese references and averaged ECAPA-TDNN/WavLM-large cosine similarities [15, 16]. Lao uses 404 FLEURS [17] rows for 260 inputs, one reference, and Gemini scoring; CER comparisons are within-language. Common400 reuses FLEURS train/development conditions. Normalized text checks find no overlap with archived DVB transcripts or between Lao S/R and evaluation inputs; Burmese hashes cover only 100 target/reference conditions without matches. Audio near-duplicate and speaker-overlap audits remain incomplete. FireRedTTS3 decoding uses seed 42, 10 flow steps, guidance 2.0, and stop threshold 0.5; long/empty outputs remain scored. Content–speaker balance is $H = 200as/(a + s)$, where $a = \text{clip}(1 - \text{CER}/100, 0, 1)$, $s = \text{clip}(\text{SIM-O}, 0, 1)$, and $H = 0$ if $a + s = 0$. Naturalness MOS (1–5) averages 75 crossed ratings from five listeners on all 15 clips per system; no confidence intervals are reported.

3.2. Main results

Table 1 shows order dependence: uniform S→R restores similarity but can worsen CER; R→S favors CER over similarity. Cubic ranks highest in H and reported MOS across all three groups. Rankings are descriptive (OmniVoice’s H margin over R: 0.08/100). Matched-batch Lao shows the same trade-off.

Adapted OmniVoice leads the shared Burmese comparison (Table 2), reducing Base CER by 8.19 points. Fixed-voice SIM-O is descriptive, with identical input texts. All outputs remain scored: mmSpeech Tacotron hits its 12.54 s limit on 210/400 cases, and IMS-Toucan emits Burmese phoneme warnings.

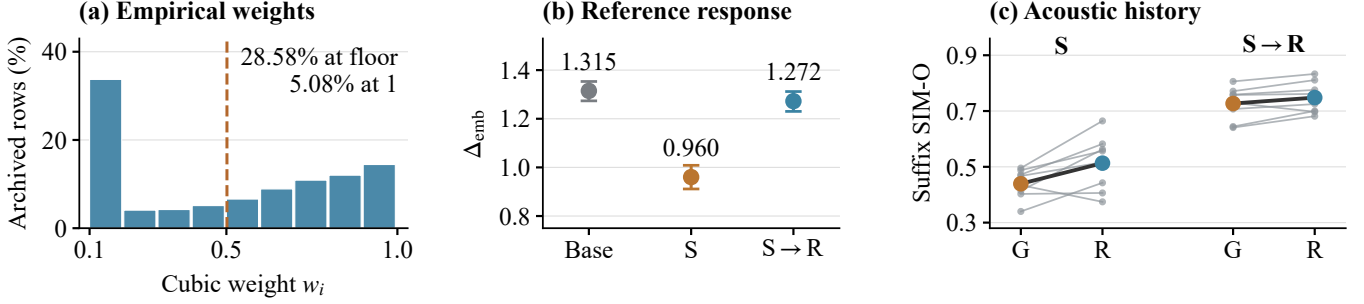


Fig. 2. (a) Cubic weights on the 42,785-row ASR archive. (b) Embedding response across 16 texts; paired-bootstrap 95% intervals. (c) Eight suffix SIM-O pairs: generated (G) versus recorded (R) prefixes; thick lines: means. Panels (b,c) use historical uniform 0.5 S→R, not cubic weighting.

Table 3. Burmese FireRedTTS3 controls from S (Common400/Clone300). (a) Same FLEURS recordings, alternative labels. (b) Matched broadcast weighting runs; permutations averaged within seed.

Setting	CER (%) ↓	SIM-O ↑
<i>(a) Training-label control</i>		
S starting point	17.43	0.4513
ASR pseudo-labels, $w = 0.5$	21.17	0.6345
ASR pseudo-labels, $w = 1.0$	23.67	0.6487
FLEURS official text, $w = 1.0$	15.45	0.6342
<i>(b) Weight assignment and loss scale</i>		
Uniform 1.0	23.327	0.69391
Uniform 0.5	21.023	0.69917
Linear ($\gamma = 1$)	23.247	0.69555
Cubic, globally shuffled	21.342	0.69843
Cubic, duration-stratified shuffle	21.130	0.69881
Uniform, equal mean (0.500946)	21.086	0.69925
Cubic ($\gamma = 3$)	16.902	0.69973

4. ABLATION AND ANALYSIS

4.1. Supervision order and pseudo-label reliability

Label quality. Table 3a replaces pseudo-labels with existing FLEURS annotations on identical recordings: 3,058 (12.135 h), retaining 2,921 (11.578 h) after overlap/duration exclusions. Official labels reduce CER by 8.22 points at $w = 1$, retaining similarity gains over S; halving pseudo-label weight helps less. Label noise contributes to degradation, but other domain effects remain.

Weight assignment. Table 3b fixes initialization, targets, references, order, and updates. Global permutations (seeds 101, 202, 303 crossed with three training seeds) preserve weights but break example assignment; duration-decile shuffling controls length correlations. Both shuffles and equal-mean uniform weighting underperform cubic, supporting assignment beyond loss scale. The 42,785-row archive (Fig. 2a; mean 0.50166, SD 0.33337; quartiles 0.10000, 0.54771, 0.81090) includes 588 later-excluded rows and omits later transcripts. It differs from the 48,043-target pool (mean 0.500946), whose duration-weighted mean is 0.58715, versus 0.50110 after global and approximately 0.5865 after duration-stratified shuffling.

Mixtures. A 0/10/25/50/75/90/100% real-example sweep from S uses 16,015 updates with unit S/cubic R weights; synthetic replay restores less similarity than real-only continuation. Joint training from Base matches 160,920 S/48,045 R exposures and the learning-

rate schedule (restart at 53,640), but yields 20.160% CER/0.65120 SIM-O, supporting progressive adaptation under matched exposure.

4.2. Reference conditioning and acoustic history

Reference response. We independently swap acoustic prompts and speaker embeddings from two same-text references over 16 target texts. For generated speech g , $q = \text{SIM}(g, r_A) - \text{SIM}(g, r_B)$ uses SIM-O’s cosine similarity. The embedding-switch contrast Δ_{emb} averages the r_B -to- r_A embedding-induced change in q over both prompts and all texts (Fig. 2b). Base/S/historical uniform 0.5 S→R score 1.3147/0.9602/1.2723; matched-reference SIM-O is 0.7804/0.5672/0.7625. Conflicts follow the embedding-selected voice. R restores response weakened by S; cubic weighting is not tested here.

History dependence. Eight of 23 prefix pairs share normalized ASR text with CER $\leq 5\%$. Fixing reference, boundary, and subsequent random state, recorded versus generated prefixes improve suffix SIM-O by 0.0739/0.0211 for S/historical uniform 0.5 S→R (Fig. 2c), with 7/8 improvements each. Paired 95% intervals [0.0225, 0.1204]/[0.0025, 0.0374] condition on cases, not training seeds. Scoring excludes prefixes and the first 250 ms; unchanged-history replay reproduces outputs. This supports history dependence, not naturally accumulating voice drift.

5. CONCLUSION

We proposed synthetic-to-real adaptation with transcript-agreement weighting for low-resource TTS. The results show that supervision order and pseudo-label reliability jointly shape the balance between content accuracy and reference-speaker control. Across FireRedTTS3 on Burmese and Lao and OmniVoice on Burmese, the approach improves content accuracy and naturalness while maintaining competitive speaker similarity. Label controls identify transcription noise as a contributor to degradation, and shuffled and equal-mean weighting controls support informative example assignment beyond mean loss scale. Reference and history interventions further link the observed trade-off to changes in conditioning response, while the evidence remains specific to the evaluated settings. The central implication is that weak supervision should be organized around the capabilities it strengthens or preserves, rather than treated as interchangeable additional data. This perspective makes supervision order and reliability complementary design choices for extending zero-shot voice cloning to underrepresented languages with reduced reliance on manual transcription.

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